Control Regions with Beam-On Data

Measurement of $\nu_\mu + \bar{\nu}_\mu$ Charged-Current Single Pion Production on Argon in the NuMI Beam

First results with real data; the signal region remains blind

2026-09-09, updated 2026-09-16, 2026-10-01 and 2026-10-10

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Abstract

Update 2026-10-10. The p_μ shape failures come from the muon momentum of exiting tracks, which multiple Coulomb scattering (MCS) measures: above about 1.2 GeV/c the data hold half the predicted share of exiting muon candidates, while the MCS and range momenta of stopping muons agree within 1.5% up to 3 GeV/c. The difference is an angular noise of reconstructed tracks on the scale of the MCS segments that the simulation lacks. The simulated MCS momentum of exiting muons is calibrated to these control regions from version 1.9 of the analysis documents (§10); after the calibration the released p_μ bins agree with the data shape within 4% (the highest bin was at 0.48), and the frozen protocol with the calibrated simulation passes every shape test.

Update 2026-10-02. The pion identification of the single-pion selection no longer requires a Bragg-pion score of at least 0.08 (analysis documents, version 1.7): the beam-on, beam-off and dirt ntuples of FHC Run 1 and RHC Runs 1 and 3b do not carry that score. The control regions change with the selection, so the beam-on files were skimmed again and every comparison was repeated (§9). The frozen global test passes, $\chi^2 = 7.69/8$ ($p = 0.46$); no full-covariance pull exceeds 1 and the p_μ shape failures are unchanged. The joint fit gives CC0π at 0.991 ± 0.190 (1.021 ± 0.191 without FHC Run 1), and the FHC Run-1 residual is 0.869 ± 0.021. Where §9 gives a number, it supersedes §8 and the sections below.

Update 2026-10-01. Every prediction in this note took the dirt from the Run-1 dirt sample alone, scaled by the data POT over the POT of the neutrino overlays and with the 0.65 dirt weight applied twice, which left the dirt seven to twenty times too small (§8); the extraction never used that scaling. With each run's dirt sample scaled by the data POT over its own POT, the frozen global test passes, $\chi^2 = 6.03/8$ ($p = 0.64$; it was 8.11/8), no full-covariance pull exceeds 1 and the RHC cosmic pull falls from 1.60 to 0.18; the p_μ shape failures are unchanged. The joint fit gives CC0π at 0.975 ± 0.191 (1.013 ± 0.192 without FHC Run 1), so no correction is warranted, and the FHC Run-1 residual is unchanged, 0.862 ± 0.021. The cosmic region now agrees with the prediction (0.98), while its far versions, where the dirt is a quarter of the prediction, lie below it (0.79 and 0.76). Where §8 gives a number, it supersedes the one in the sections below.

Update 2026-09-16. Two defects in the comparison below were found while building the joint FHC+RHC background fit (§5, §6); the statements they affect are marked *superseded* where they occur. (i) **The sideband Monte Carlo was stale.** It was processed on 2026-08-11, before the beam-frame correction of the muon angle, and stores $\cos\theta_\mu$ about the detector z axis while the beam-on skim stores the beam-frame angle. **The muon-angle shape failures reported below are an artefact of that mismatch**: with the current Monte Carlo all $\cos\theta_\mu$ shape tests pass ($p = 0.24$–0.98). The p_μ shape failures in the π^0 region and in RHC CC0π remain. (ii) **The detector-variation term of the control-region covariance was inflated about tenfold.** The knob samples do not have the POT of the detector-variation central-value sample (e.g. 0.913 for WMX in FHC); the sideband macros compared unscaled counts. Normalised, the detector term on the CC0π and π^0 totals is 1–2%, not 20%. With the corrected covariance and the wrong FHC Run-1 exposure the frozen global test failed, $\chi^2 = 69.5/8$; a free FHC Run-1 exposure factor fitted $f = 0.520 \pm 0.020$, which led to the bookkeeping check that resolved it (§3). **On the corrected exposure the frozen global test passes**, $\chi^2 = 8.1/8$, $p = 0.42$, and the all-period fit needs no parameter (5.7/6). A Run-1-specific residual survives: every other period sits 12–25% above prediction, inside the correlated flux and cross-section terms, while FHC Run 1 sits at unity, so a free Run-1 factor still fits 0.869 ± 0.022 ($\Delta\chi^2 = 37$ for one parameter). It is an open item (§6). The statistical part of the detector variations is removed with an event-matched bootstrap (§5.2). A joint FHC+RHC fit of the background classes finds no normalisation correction warranted (CC0π 0.949 ± 0.190 over all periods, 0.984 ± 0.191 without FHC Run 1) and FHC and RHC consistent; its projected benefit and its limits are in §6. *All of this is post-inspection work on already-opened regions; nothing here is a frozen-procedure response or a release input.*

Original abstract (2026-09-09). The four inclusive control regions of the CC $1\mu 1\pi Xp$ NuMI analysis have been opened on beam-on data. This note reports what was processed, the yields and distributions in each region, and the statistics of the frozen comparison procedure applied to the data. The signal region has *not* been opened: the beam-on files were processed into a signal-region-stripped skim, verified to contain no event passing the signal selection.

Two things must be settled before any of these numbers is read as a statement about the background model. First, **no Run-2 beam-on file exists**, so Run 2 is excluded from the data *and* from the prediction throughout; the comparison covers 85.7% of the FHC and 76.6% of the RHC exposure. Second, and more seriously, the run-by-run comparison is not uniform: **FHC Run 1 sits at a data/prediction ratio of** 0.64–0.72 **in the high-statistics regions while every other run period of either beam mode lies between** 1.08 **and** 1.25. Flux and detector uncertainties are common to the run periods of a mode and cancel in that comparison, so the difference is not a systematic of the prediction. The natural explanation is the exposure assumed for the staged FHC Run-1 beam-on file, which no quantity in this comparison verifies. Until that is resolved the mode-level ratios of §2 should not be interpreted.

With FHC Run 1 set aside, the remaining five run periods sit coherently 10–25% above the prediction in all four regions. That is a common normalisation offset, the one direction the frozen global test is explicitly insensitive to, and it is within the 20–30% prediction uncertainty. It is nonetheless coherent across regions of very different composition, and is the second thing to understand.

Applied to the data, the frozen procedure's pass/fail statistic — the global normalisation

χ^2 over the eight (region, mode) totals — gives $8.77/8$, $p = 0.362$, and no region is flagged by the per-region pulls. The shape tests, which project the normalisation out and are therefore blind to all of the above, are a different matter: the muon-angle distribution fails in the CC0π and π^0 regions in *both* beam modes, by a wide margin, while the sixteen other tests are unremarkable. The same code on pseudo-data returned all twenty-two with $p > 0.47$. Under the frozen rules that removes both of those regions from eligibility for a fitted normalisation correction, which requires a passing shape test.

Contents

1 What was opened, and what was not **2**

2 Yields and distributions **3**

3 The run-by-run comparison **4**

4 The frozen comparison procedure applied to the data **5**

5 Corrections found on 2026-09-16 **7**
5.1 The sideband Monte Carlo predates the beam-frame correction 7
5.2 Detector variations: POT normalisation and the statistical part 8
5.3 The frozen procedure with corrected inputs 9

6 Joint FHC+RHC background fit **10**
6.1 Model 10
6.2 What two beams buy 11
6.3 Expected effect on the measurement 11
6.4 Validation 12
6.5 Application to the beam-on control regions 13

7 Far sidebands: control regions with less signal in them **14**

8 Correction found on 2026-10-01: the dirt normalisation **16**

9 Selection change of 2026-10-02: the Bragg-pion requirement **18**

10 MCS momentum of exiting muons **20**

11 What this establishes, and what it does not **22**

1 What was opened, and what was not

The analysis is blind in the signal region. Beam-on data were processed with the control-region skim: an event is written only if it *fails* the complete signal-region selection *and* satisfies at least one of the four approved control-region definitions. The written file carries a marker that the cross-section extraction refuses, and the sideband tools accept a beam-on file only if it carries that marker. Each processing job re-read its own output and deleted it unless the marker was present, no written event carried the signal-region flag, and every written event fell in at least one control region. All eight files passed.

Because the skim keeps every control-region event and discards only events that are in the signal region or in no region at all, the control-region distributions below are complete: nothing that belongs in a control region has been dropped.

Table 1: Beam-on files processed. "Input events" is the number of entries in the PeLEE ntuple; the run range is read from the file and is what fixes the horn-mode assignment of the Run-4 sub-periods, which the file names do not always carry. The beam-off files of the same production have the mode in their names and the same run ranges, which is how 4a/4b are identified as RHC and 4c/4d as FHC.

Period	File	Run range	Input events
FHC Run 1	`run1_beamon_beamgood`	4952–6748	610 496
FHC Run 4c	`beam_on_run4c`	21286–22269	392 148
FHC Run 4d	`beam_on_run4d_fhc`	22270–23259	228 821
FHC Run 5	`beam_on_run5`	24376–25759	748 669
RHC Run 1	`run1_RHC_beamon_beamgood`	6748–6998	167 923
RHC Run 3	`run3b_beamon_beamgood`	14117–17566	1 104 349
RHC Run 4a	`beam_on_run4a`	19660–19752	43 027
RHC Run 4b	`beam_on_run4b`	19753–21285	835 573

The CRT-era trigger flag. The Run-4 and Run-5 beam-on files carry a software-trigger flag that is identically zero and no companion pre-CRT flag, exactly the pattern of the CRT-era beam-off samples. The branch is bound only when the companion branch is present, so these files keep the fail-open default and are not rejected. Had that not been the case, Runs 4 and 5 would have processed to zero events silently. This was verified on the beam-on files directly before processing.

No Run-2 beam-on data. The staged files cover run numbers 4952–6998, 14117–17566 and 19660–25759. Nothing lies in the range 6999–14116, which contains all of Run 2. The exposure the prediction is normalised to assigns Run 2 1.268×10^{20} POT in FHC and 2.591×10^{20} in RHC, so the missing period is 14.3% of the FHC, 23.4% of the RHC and 19.4% of the combined exposure. Run 2 is therefore dropped from the data and from the prediction — the Monte Carlo, the beam-off and the dirt — everywhere in this note. The beam-off side of the analysis stands Run 1 in for Run 2 permanently, because beam-off is a cosmic-only sample matched by run period; that substitution is not available for beam-on data, which is the measurement.

Table 2: Exposure entering this comparison, against the full exposure the analysis is normalised to.

	Runs used	POT used (10^{20})	Fraction of the mode
FHC	1, 4, 5	7.589	85.7%
RHC	1, 3, 4	8.491	76.6%

2 Yields and distributions

The prediction is the same stack as the blind studies: neutrino Monte Carlo scaled per run to that run's data exposure with the central-value weight, plus the per-run beam-off sample scaled by its gate ratio, plus the dirt overlay. Unlike the pseudo-data studies, *nothing is added to the data*: real beam-on data already contains cosmics and dirt, which here belong to the prediction alone.

Table 3: Control-region yields with beam-on data, summed over the run periods used. The prediction components are the neutrino signal contamination, the neutrino background, the beam-off and the dirt. **The mode-level ratios in the last column mix the FHC Run-1 anomaly of §3 with the rest and should not be read on their own.**

Mode	Region	Data	Prediction	signal	ν bkg	beam-off	dirt
FHC	CC0π	10 660	11 571.0	962.3	9 442.8	1 118.1	47.8
	multi-π	109	109.6	24.3	79.6	5.7	0.1
	π^0	6 291	6 327.3	562.2	5 119.5	635.4	10.1
	cosmic	92	108.3	5.0	59.5	42.0	1.9
RHC	CC0π	12 047	10 558.9	915.2	8 763.9	846.7	33.1
	multi-π	136	125.0	27.1	94.3	3.6	0.1
	π^0	8 092	6 639.5	594.0	5 578.8	459.7	7.0
	cosmic	120	89.5	5.0	58.0	25.2	1.3

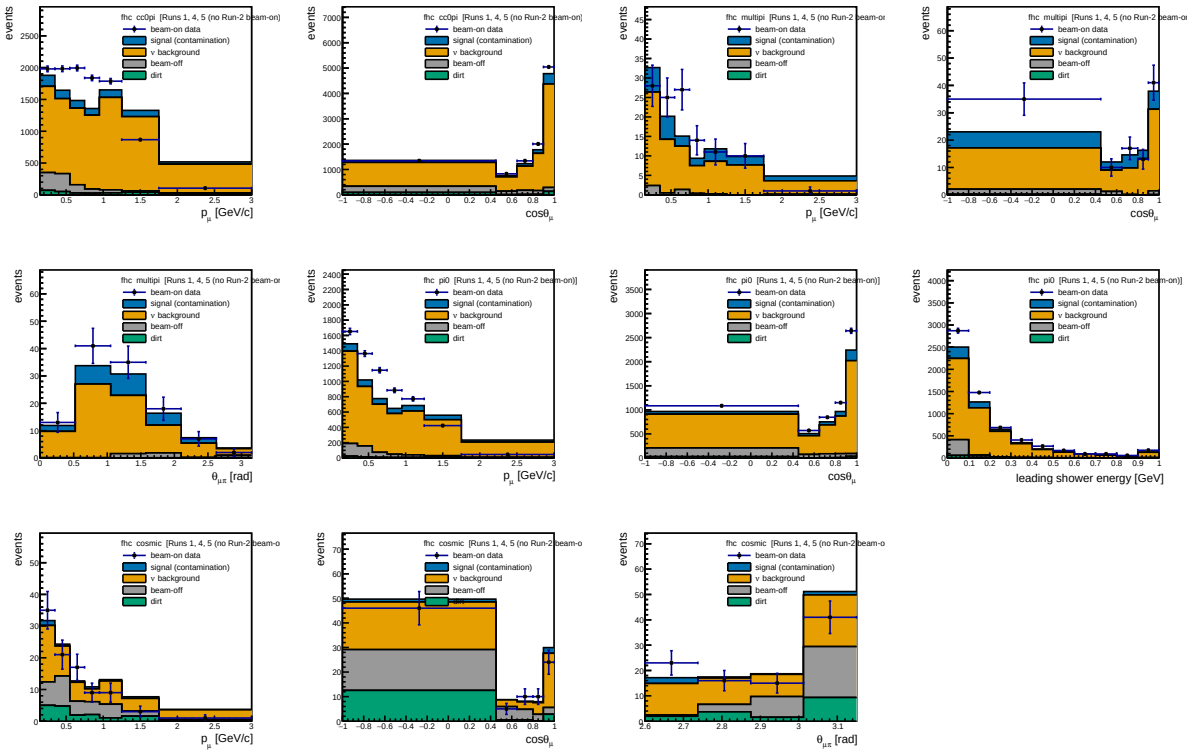


Figure 1: FHC control regions with beam-on data (Runs 1, 4, 5). Rows are the CC0π, multi-π, π^0 and cosmic regions; the distributions are the frozen ones. Points are beam-on data with Poisson errors; the stack is the prediction. Regenerated 2026-09-16 from the current sideband Monte Carlo (§5.1).

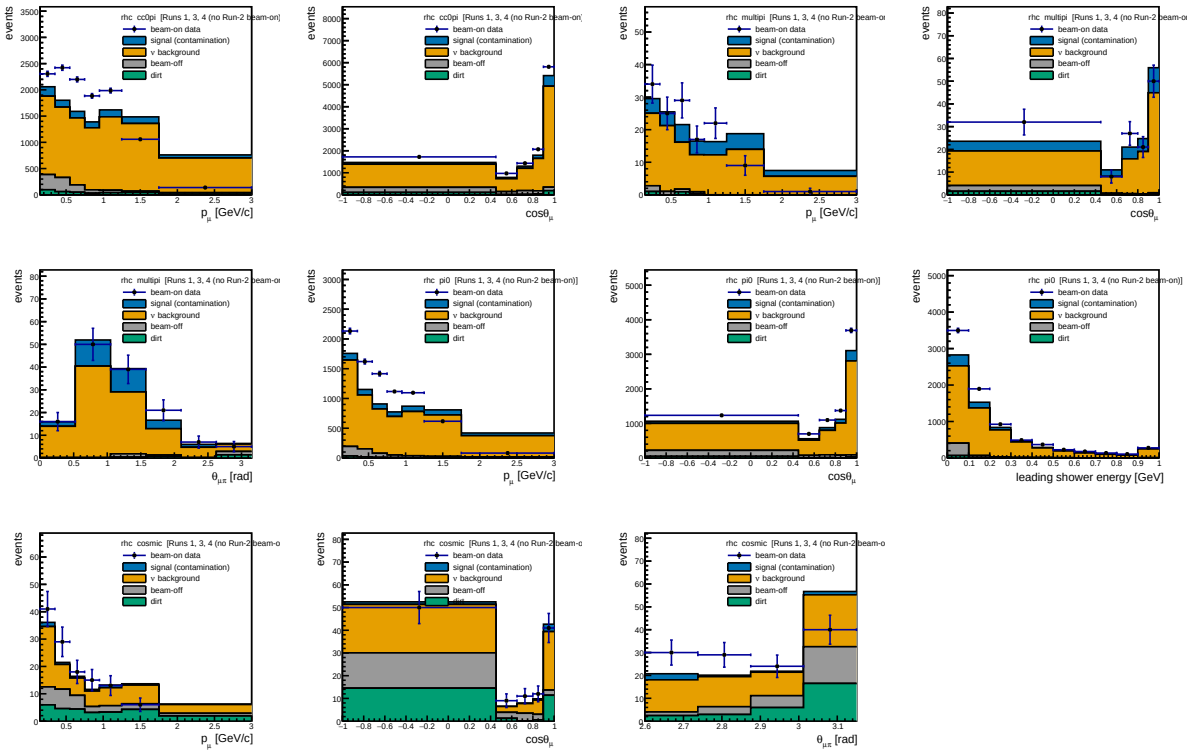


Figure 2: RHC control regions with beam-on data (Runs 1, 3, 4). Same layout as Fig. 1; regenerated 2026-09-16 from the current sideband Monte Carlo.

3 The run-by-run comparison

The frozen procedure asks for the comparison run period by run period, and it is what makes the mode-level numbers of §2 uninterpretable as they stand.

Table 4: Data over prediction, per run period and region. Each period's prediction uses that period's Monte Carlo scaled to its own exposure, its own beam-off sample and gate ratio, and the dirt scaled to its exposure. FHC Run 1 is the outlier; every other period of either mode lies between 0.83 and 1.51, with the high-statistics regions between 1.08 and 1.25.

Period	POT (10^{20})	CC0π	multi-π	π^0	cosmic
FHC Run 1	3.283	**0.637**	0.911	**0.715**	**0.595**
FHC Run 4	2.075	1.173	1.217	1.248	1.141
FHC Run 5	2.231	1.135	0.899	1.188	1.200
RHC Run 1	0.605	1.191	1.076	1.086	0.828
RHC Run 3	5.003	1.150	1.243	1.246	1.510
RHC Run 4	2.883	1.115	0.823	1.201	1.248

FHC Run 1 is not a systematic of the prediction. The flux uncertainty is a hadron-production model uncertainty common to every run period of a beam mode, and the detector variations are taken from the same Run-4 samples for all periods, so both are almost fully correlated between run periods and cancel in a period-to-period comparison. A prediction systematic can move all FHC periods together; it cannot put Run 1 at 0.64 while Runs 4 and 5 sit at 1.17 and 1.14. The effect is also flat across the four regions of Run 1 — 0.64, 0.91, 0.72, 0.60 — which is the signature of a normalisation applied to the whole period rather than of a

mis-modelled background class, since the four regions have very different compositions.

The quantity that is not verified anywhere in this comparison is the Run-1 FHC beam-on exposure itself. The prediction is scaled to 3.283×10^{20} POT and the beam-off to 9 846 635 beam-on gates, both taken from the trigger table; the beam-on ntuples carry no exposure record of their own, so nothing here cross-checks them against the file that was actually staged. The staged FHC Run-1 file spans runs 4952–6748 and the RHC Run-1 file begins at run 6748, so Run 1 is split between the horn modes inside the run range, which is exactly the situation in which a partial file or a mis-assigned split is easy to miss. The size of the effect is a concrete, falsifiable prediction. Every component of the Run-1 prediction — the Monte Carlo through its POT scale, the beam-off through its gate ratio, the dirt through its exposure — is proportional to the assumed Run-1 exposure, so an exposure error rescales the whole prediction. Bringing FHC Run 1 onto the 1.15–1.20 of the other periods requires the true exposure to be a fraction $f \simeq 0.51$–0.62 of the assumed one, that is roughly 1.7–2.0×10^{20} POT in place of 3.283×10^{20}, with the three high-statistics regions giving $f = 0.55$ (CC0π), 0.62 (π^0) and 0.51 (cosmic). The consistency of f across regions of different composition is itself the evidence that a single multiplicative exposure factor, and not a background class, is what is wrong. If instead the trigger table is confirmed against the staged file, the deficit is real and must be treated as a localised reconstruction failure under the response tree.

The remaining five periods carry a common offset. Setting FHC Run 1 aside, the five other periods sit above the prediction by 10–25% in the high-statistics regions, in both beam modes. This is a common normalisation offset. It is within the 20–30% prediction uncertainty, and it is the one direction the global test of §4 is built to be insensitive to, because the correlated flux, cross-section and detector terms absorb it. Two features are worth recording for whatever investigation follows: the offset appears in regions whose beam-off fraction ranges from 3–5% (multi-π) to 28–39% (cosmic), which argues against a beam-off scaling error; and it appears in both horn modes, which argues against a mode-specific flux problem.

4 The frozen comparison procedure applied to the data

The frozen procedure defines one pass/fail statistic — the global normalisation χ^2 over the eight (region, mode) totals with the full pre-fit covariance — and two diagnostics, the per-region pulls and the normalisation-marginalised shape tests of the frozen distributions. All three are computed here with exactly the machinery that was exercised on pseudo-data before opening, with only the data, the retained run periods and the removal of the pseudo-data beam-off throw changed. On pseudo-data that machinery returned a global χ^2 of $0.66/8$ and all twenty-two shape tests with $p > 0.47$, so it is not predisposed to fail.

Table 5: Per-region normalisation pulls and the global test. "full" is the pre-fit prediction covariance plus the data Poisson term; "st+det" is the statistical and detector terms alone. The last column is the fractional shift of that region's total which, on its own, would bring the global χ^2 to its 95% point.

Mode	Region	Data	Prediction	EXT	pull (full)	pull (st+det)	σ_{full}/N
FHC	CC0π	10 660	11 571.0	9.7%	−0.29	−0.40	27.0%
	multi-π	109	109.6	5.2%	−0.02	−0.03	34.2%
	π^0	6 291	6 327.3	10.0%	−0.02	−0.03	29.6%
	cosmic	92	108.3	38.7%	−0.67	−0.80	22.4%
RHC	CC0π	12 047	10 558.9	8.0%	0.73	1.75	19.3%
	multi-π	136	125.0	2.9%	0.26	0.41	33.1%
	π^0	8 092	6 639.5	6.9%	0.99	3.14	22.1%
	cosmic	120	89.5	28.2%	1.63	2.21	20.9%
Global normalisation $\chi^2 = 8.77/8$, $p = 0.362$ — **passes** ($p > 0.05$ required)							
Blind spot: a common shift of all eight totals fails the test only beyond 61.9%							

Normalisation: the set passes. The global χ^2 is 8.77 on eight degrees of freedom, $p = 0.362$. No region is flagged by the $|\text{pull}| > 2$ criterion against the full covariance; the largest is the RHC cosmic region at 1.63. That is the expected outcome given §3: the offsets are 10–25% against a prediction uncertainty of 19–34% per region, and they are strongly correlated between regions (0.4–0.9), which is exactly the direction the test cannot resolve. Against the statistical and detector terms alone, which is the diagnostic column, RHC π^0 reaches 3.14 and RHC cosmic 2.21.

Table 6: Normalisation-marginalised shape tests of the frozen distributions. The overall normalisation of each distribution is projected out before the χ^2 is formed, so these are insensitive to the offsets of §3. Entries with $p < 0.01$ in bold. **Superseded 2026-09-16**: the prediction used here came from the stale sideband Monte Carlo (§5.1); corrected values in Table 9.

Mode	Distribution	χ^2/ndf	p	Mode	Distribution	χ^2/ndf	p
FHC	CC0π p_μ	13.09/6	0.042	RHC	CC0π p_μ	**52.85/6**	**0.000**
	CC0π $\cos\theta_\mu$	**350.51/4**	**0.000**		CC0π $\cos\theta_\mu$	**583.58/4**	**0.000**
	multi-π p_μ	6.29/6	0.391		multi-π p_μ	10.67/6	0.099
	multi-π $\cos\theta_\mu$	8.49/4	0.075		multi-π $\cos\theta_\mu$	9.97/4	0.041
	multi-π $\theta_{\mu\pi}$	0.75/5	0.980		multi-π $\theta_{\mu\pi}$	0.64/5	0.986
	π^0 p_μ	**30.41/6**	**0.000**		π^0 p_μ	**67.40/6**	**0.000**
	π^0 $\cos\theta_\mu$	**210.36/4**	**0.000**		π^0 $\cos\theta_\mu$	**240.85/4**	**0.000**
	π^0 E_{shr}	7.14/9	0.623		π^0 E_{shr}	7.20/9	0.616
	cosmic p_μ	4.54/6	0.604		cosmic p_μ	4.22/6	0.647
	cosmic $\cos\theta_\mu$	5.56/4	0.234		cosmic $\cos\theta_\mu$	4.27/4	0.371
	cosmic $\theta_{\mu\pi}$	2.80/3	0.423		cosmic $\theta_{\mu\pi}$	2.12/3	0.547

Shape: the muon angle fails, and it is not the normalisation offset. ***Superseded 2026-09-16: the muon-angle failures are an artefact of the stale sideband Monte Carlo (§5.1); the corrected tests are in Table 9.*** The shape tests project the overall normalisation out, so they are blind to everything §3 is about. Six of the twenty-two fail at $p < 0.01$, and they are not scattered: they are the muon momentum and, far more strongly, the muon angle in the two high-statistics regions, CC0π and π^0, in *both* beam modes. The $\cos\theta_\mu$ values, 351/4 and 210/4 in FHC and 584/4 and 241/4 in RHC, are not marginal. The remaining sixteen tests, including every distribution of the multi-π and cosmic regions and the π^0 shower energy that the region was built to test, are unremarkable.

Four things bound how far this can be taken.

- **It is not an artefact of the test.** The same code on pseudo-data returned all twenty-two with $p > 0.47$. The failures appear only when real data is substituted.
- **It is not the FHC Run-1 exposure.** A wrong weight on one run period does distort a summed distribution, because the periods do not have identical shapes, so the FHC numbers are not clean. But the largest failures are in RHC, which contains no period suspected of an exposure error.
- **The test is very sensitive by construction.** With 10^4 events per region and a covariance dominated by normalisation-like flux and detector terms, projecting the normalisation out leaves a small shape covariance, so a discrepancy of a few per cent per bin produces a large χ^2. The size of these numbers should not be read as the size of the effect.
- **The binning concentrates the forward peak.** The released $\cos\theta_\mu$ edges are -1, 0.45, 0.65, 0.8, 0.9, 1: the first bin spans three quarters of the range, so the test is largely comparing the forward peak with everything else. A forward-angle discrepancy is where the MicroBooNE BNB CC1$\pi^\pm$ measurement also found every tested generator above the data.

Procedural consequence. *(Superseded 2026-09-16 for the muon angle; the p_μ failures of the π^0 region and of RHC CC0π survive the correction, so the conclusion on eligibility for a fitted correction is unchanged.)* Under the frozen procedure a fitted normalisation correction is permitted only for a region whose shape test passes, so that the discrepancy is consistent with a pure normalisation shift. On these numbers neither CC0π nor π^0 qualifies: both fail their muon-angle shape test in both modes. Whatever the eventual explanation, a fitted correction to either region is not available under the rules as frozen, and the response tree points instead at the added-uncertainty branch or at a model revision with renewed approval. That conclusion should be revisited once the exposure questions of §3 are closed and the tests are repeated.

5 Corrections found on 2026-09-16

Both defects were found while building the templates of the joint background fit of §6, whose first version reproduced the frozen statistics exactly — including their failures — and so inherited them.

5.1 The sideband Monte Carlo predates the beam-frame correction

The reconstructed muon angle of the NuMI analysis is measured about the neutrino direction, approximated by the vector from the NuMI target to the reconstructed vertex; before 2026-08-19 it was measured about the detector z axis, 28° away. The sideband-instrumented Monte Carlo used for every figure and table of this note (`processed/sb/`) was produced on 2026-08-11. Recomputing the angle from the stored track direction on 3 000 CC0π events of each file settles which convention each file carries (Table 7): the sideband Monte Carlo stores the detector-frame angle, the beam-on skim and every current Monte Carlo tree the beam-frame angle. The totals are unaffected — the region flags do not depend on the angle convention — so the yields, the per-run table and the global normalisation test are unchanged by this defect; every $\cos\theta_\mu$ distribution and shape test is not.

Table 7: Left: median $|\cos\theta_\mu^{\text{stored}} - \cos\theta_\mu^{\text{recomputed}}|$ for the two conventions. Right: data over prediction in the CC0π region, FHC Runs 4+5, in bins of reconstructed $\cos\theta_\mu$ (edges $-1, 0, 0.45, 0.65, 0.8, 0.9, 0.95, 1$), with the prediction from the stale and from the current Monte Carlo, and with the angle recomputed in the detector frame for both (identical by construction).

File	vs. beam frame	vs. detector z
`sb/` (2026-08-11)	0.155	0.000
`processed/` (2026-08-19)	0.000	0.155
`sb_pi0/` (2026-09-06)	0.000	0.154
beam-on skim (2026-09-09)	0.000	0.163

	$\cos\theta_\mu$ bin						
Prediction	1	2	3	4	5	6	7
`sb/`, beam frame	1.33	1.09	0.73	0.71	1.00	1.28	2.22
`sb_pi0/`, beam frame	1.34	1.10	1.19	1.18	1.21	1.13	1.13
detector frame	1.38	1.38	1.21	1.12	1.07	1.10	1.24
$p_\mu > 0.75$ GeV/c							
`sb/`, beam frame	1.13	0.38	0.22	0.36	0.67	1.15	2.46
`sb_pi0/`, beam frame	1.26	0.89	0.80	0.96	1.03	0.98	1.04

The swing of the stale comparison from 0.22 to 2.46 disappears with a consistent convention: the current prediction is flat at the 1.10–1.34 offset of §3. All sideband macros now read `processed/sb_pi0/` (reprocessed 2026-09-06 with the current selection; it also carries the Run-4 detector-variation samples, the per-run beam-off and the dirt). Figures 1 and 2 have been regenerated from it; the originals are kept as `sideband_data_*_sb20260811.pdf`.

5.2 Detector variations: POT normalisation and the statistical part

The sideband macros formed the detector-variation shifts as ratios of unscaled event counts, on the grounds that "the POT scale cancels in every ratio". It does for a transfer factor $N^{\text{SR}}/N^{\text{CR}}$ within one sample, so the transfer-factor uncertainties of the analysis note stand; it does not for a knob-to-central-value ratio, because the knob samples were not produced with the central value's POT (Table 8). The cross-section extraction normalises by `summed_pot` and is not affected.

Table 8: FHC Run-4 detector variations: POT of each knob sample relative to the central-value sample, and the shift of the region totals (%) as unscaled count ratios (u) and POT-normalised with the event-matched bootstrap noise (n). The RHC samples show the same pattern with POT ratios of 0.950–1.012.

		CC0π		π^0		signal region	
Knob	POT ratio	u	n	u	n	u	n
LYdown	0.926	-8.1	-0.8 ± 0.3	-7.6	-0.2 ± 0.3	-7.0	$+0.5 \pm 0.6$
LYrayl	0.916	-8.1	$+0.4 \pm 0.3$	-8.0	$+0.4 \pm 0.3$	-7.1	$+1.4 \pm 0.5$
Recomb2	1.039	$+4.4$	$+0.8 \pm 0.4$	$+3.2$	-0.1 ± 0.5	-3.2	-6.8 ± 0.9
SCE	0.999	$+1.5$	$+1.7 \pm 0.5$	-0.7	-0.6 ± 0.6	-0.5	-0.4 ± 1.2
WMAngleXZ	0.907	-9.6	-0.4 ± 0.4	-9.2	$+0.0 \pm 0.5$	-6.8	$+2.7 \pm 0.9$
WMAngleYZ	0.911	-9.3	-0.5 ± 0.4	-9.0	-0.3 ± 0.5	-6.6	$+2.5 \pm 1.0$
WMX	0.913	-8.5	$+0.3 \pm 0.4$	-9.7	-0.9 ± 0.5	-12.1	-3.7 ± 1.0
WMYZ	0.917	-8.5	-0.3 ± 0.4	-9.1	-0.7 ± 0.5	-9.1	-0.9 ± 0.9

Removing the statistical part. A knob-minus-central-value difference also carries the statistical fluctuation of the finite samples, which a covariance built from the shifts counts as detector uncertainty. The knob samples are re-simulations of the *same* generated events: matching on

the true neutrino energy and vertex pairs 96.8–97.0% of the FHC knob events with central-value events (94.1% for Recomb2) and 97.6–98.1% in RHC. The statistical noise of the difference therefore comes only from events that migrate between bins and from the unmatched few per cent. It is estimated with a Poisson bootstrap in which every generated event receives one Poisson(1) weight shared by its central-value and knob copies (300 replicas):

$$f_{k,i} = \frac{N_i^k \,\mathrm{POT_{CV}}/\mathrm{POT}_k - N_i^{\mathrm{CV}}}{N_i^{\mathrm{CV}}}, \qquad C_k = \mathrm{Cov}_{\mathrm{bootstrap}}(f_k), \tag{1}$$

$$V_{\mathrm{det},ij} = \Big[\textstyle\sum_k \left(f_k f_k^{\mathsf{T}} - C_k\right)\Big]_{\mathrm{PSD},\,ij} \mu_i\,\mu_j, \tag{2}$$

since $\mathrm{E}[f f^{\mathsf{T}}] = \varphi\varphi^{\mathsf{T}} + C$ for true shifts φ; the sum is projected onto the positive semi-definite cone by eigenvalue clipping. On the CC0π total the bootstrap noise is 0.3% against 0.9% for independent samples. On region totals the subtraction is a small correction; on finely binned distributions it is not: in the $(\cos\theta_\mu \times p_\mu)$ bins of §6 the median quadrature shift is 8.3% (FHC), of which 7.3% is statistical, leaving 3.9% (Fig. 3). The resulting detector term is 1.9% on the FHC CC0π total (20.2% unscaled), 0.8% on π^0 (20.1%), 6.5% on cosmic (21.7%) and 8.3% in the signal region (21.0%); in RHC 2.1, 1.8, 7.0 and 8.5% (unscaled 8.4, 7.2, 30.9, 15.4%).

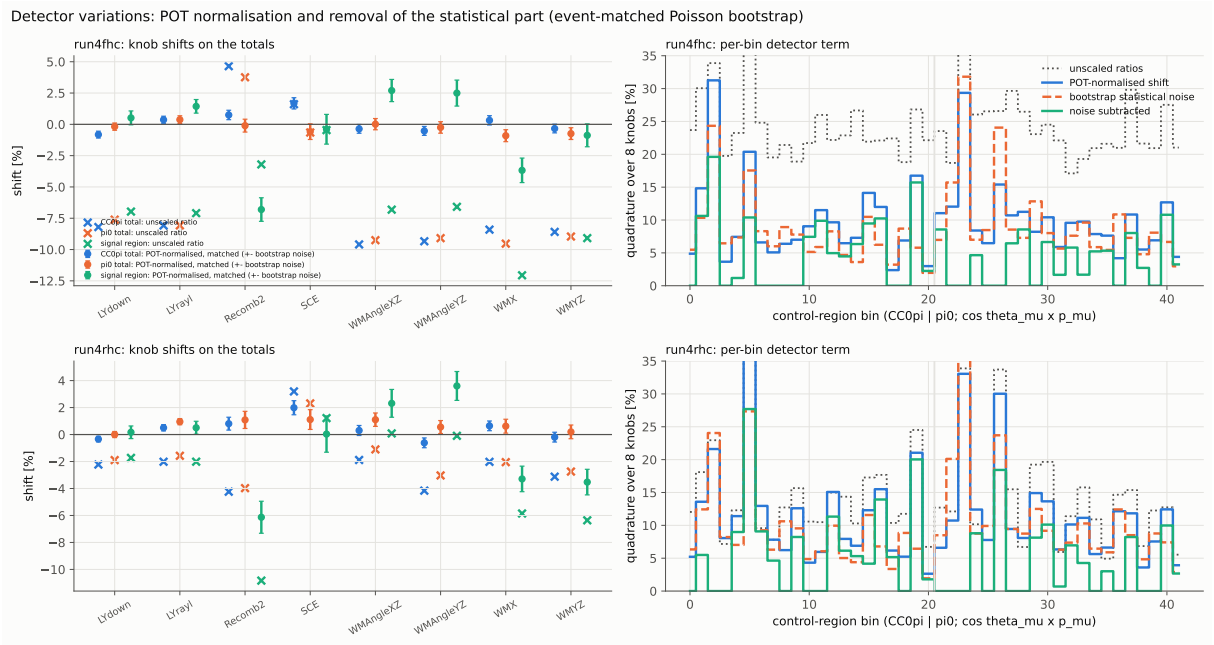


Figure 3: Left: knob shifts of the region totals, unscaled (crosses) and POT-normalised with the bootstrap noise (points). Right: quadrature over the eight knobs per $(\cos\theta_\mu \times p_\mu)$ bin of the CC0π and π^0 regions: unscaled, POT-normalised, its statistical noise, and the noise-subtracted estimate.

5.3 The frozen procedure with corrected inputs

`macros/sb_protocol_data.C` now takes two switches, current Monte Carlo and POT-normalised detector variations, both on by default; with both off it reproduces Tables 5 and 6 exactly. Its definitions are unchanged (multisim families and the eight knobs as unisims; the event-matched noise subtraction is not part of it).

Table 9: Frozen statistics with corrected inputs (current Monte Carlo, POT-normalised detector variations, corrected FHC Run-1 exposure), same run periods as Table 5. Shape tests: χ^2/ndf (p); $p < 0.01$ in bold. The RHC rows are unchanged by the exposure correction.

Mode	Region	pull (full)	pull (st+det)	σ_{full}/N	p_μ shape	$\cos\theta_\mu$ shape
FHC	CC0π	0.47	4.17	18.8%	12.72/6 (0.048)	2.59/4 (0.628)
	multi-π	0.46	0.82	33.3%	6.33/6 (0.387)	1.72/4 (0.787)
	π^0	0.77	7.66	22.6%	**28.15/6 (< 0.001)**	0.41/4 (0.981)
	cosmic	0.37	0.49	19.8%	4.22/6 (0.647)	2.45/4 (0.654)
RHC	CC0π	0.80	6.03	17.7%	**50.19/6 (< 0.001)**	5.48/4 (0.242)
	multi-π	0.28	0.47	31.5%	10.40/6 (0.109)	2.52/4 (0.640)
	π^0	1.03	9.53	21.1%	**66.61/6 (< 0.001)**	0.49/4 (0.975)
	cosmic	1.60	2.14	21.3%	4.10/6 (0.663)	0.11/4 (0.999)

Global normalisation $\chi^2 = 8.11/8$, $p = 0.42$ — **passes**; blind spot of a common shift 58.3%.
(On the wrong FHC Run-1 exposure this was 69.49/8, $p < 10^{-3}$, and the FHC pulls were -0.42, -0.02, -0.03, -0.82.)
Other shape tests all pass: multi-π $\theta_{\mu\pi}$ $p = 0.99/0.99$, cosmic $\theta_{\mu\pi}$ 0.57/0.53, π^0 E_{shr} 0.65/0.63.

Three consequences. **(1)** With the corrected covariance but the wrong FHC Run-1 exposure the global test failed (69.49/8); the fit framework of §6, which reproduces the frozen statistic (8.82/8 against 8.77/8, same pulls to 0.03), gave 48.4/8 with all periods and 5.0/8 ($p = 0.76$) without FHC Run 1, which isolated the FHC Run-1 excursion once it was no longer absorbed by a 20% detector term and led to the bookkeeping check that resolved it. On the corrected exposure the test passes, 8.11/8 ($p = 0.42$), with every full-covariance pull positive (data above prediction in all eight totals) and none beyond 1.6. **(2)** No region is flagged by the full-covariance pulls ($|\text{pull}| \leq 1.60$), but the statistical-plus-detector pulls of the RHC CC0π and π^0 regions are 6.0 and 9.5: the common offset is highly significant against everything except the flux and cross-section terms. **(3)** The muon-angle failures are gone; the p_μ failures of the π^0 region (both modes) and of RHC CC0π remain and are unexplained. Under the frozen rules the response tree is entered through item 4(i) for FHC Run 1 and neither CC0π (RHC p_μ) nor π^0 is eligible for a fitted correction.

6 Joint FHC+RHC background fit

Status. This section develops and validates the fit proposed in `BACKGROUND_FIT_PLAN.md` and applies it, *post-inspection*, to the control regions opened in this note. It is not a response under the frozen procedure, it does not touch the signal region, and none of its outputs is a release input; using any of them would be a documented amendment requiring renewed approval. Code: `xsec_analyzer/bkgfit/` and `macros/bkgfit_templates.C`; results `logs/bkgfit/*.json`.

6.1 Model

- **Templates** per run period (FHC Runs 1, 2, 4, 5; RHC Runs 1, 2, 3, 4; Run 2 Monte Carlo only, for full-exposure projections) from `sb_pi0/`: the neutrino Monte Carlo split into five truth classes (signal; ν_μCC 0π; CC+NC with a π^0; ν_μCC $\geq 2\pi^\pm$ without π^0; other) and by $\nu/\bar{\nu}$, with its central value, $\sum w^2$ and all 2 230 systematic universes (PPFX 600, GENIE multisim 600, re-interaction 1 000, the nine GENIE unisims and the two SCC pairs); the beam-off per run period with the gate ratios of the frozen procedure; the dirt; the beam-on skim.
- **Bins**: CC0π and π^0 in reconstructed $\cos\theta_\mu$ (edges $-1, 0, 0.45, 0.65, 0.8, 0.9, 0.95, 1$) $\times$ p_μ ($0.15, 0.35, 0.75, \infty$ GeV/c), 21 each, and the cosmic total ("2D"); or the three totals ("totals"). Multi-π overlaps π^0 and is left out. The binning requires $p_\mu \geq 0.15$ GeV/c, which

removes about 2% of the region events in data and prediction alike.

- **Parameters**: normalisations of truth classes, applied in every universe so that their correlation with the flux and cross-section terms is kept; optionally per beam mode, and an exposure factor per run period.
- **Covariance**: the universe families with the extraction's conventions (multisims averaged, unisims summed), the event-matched noise-subtracted detector term of §5.2 (fully correlated between run periods and between the two modes, noise within a mode), POT 2% and targets 1%, Monte Carlo, beam-off and dirt statistics, and the data term in the Pearson form (prediction variance).
- **Test statistic**: $\chi^2(\theta) = r^\mathsf{T} V^{-1} r$ minimised with Minuit2 (MIGRAD, HESSE, MINOS). Two choices are forced by validation. The systematic covariance is held at the nominal prediction: evaluated at θ, a poor fit is "improved" by inflating it and the parameters run to their limits. The data variance is held there too: evaluated at θ it grows with the prediction and biased the 2D toys upward by one standard deviation. It is the Pearson form, the prediction itself: the combined Neyman–Pearson form depends on the data and leaves a residual $+0.02$ to $+0.03\sigma$ pull bias that only 10^5-toy ensembles resolve (Sec. 13 of the analysis note). An iterated variant re-evaluates both at the result.

6.2 What two beams buy

The antineutrino share of the neutrino Monte Carlo (full exposure) is 11.8% (FHC) and 15.4% (RHC) in the CC0π region, 12.9 and 12.3% in π^0, 15.6 and 18.7% in the signal region. The NuMI RHC flux at MicroBooNE carries a large ν_μ component, and antineutrino cross sections and efficiencies are lower, so the two modes do not separate ν from $\bar{\nu}$ backgrounds: fitted jointly, a $\bar{\nu}$ CC0π normalisation is determined to ± 0.60 against ± 0.14 for ν (Table 10). What the joint fit does buy is statistics, the flux correlation between the modes and a consistency test between them.

Table 10: Expected uncertainty (Asimov, 2D bins, full exposure) of each parameter set, FHC alone / RHC alone / joint, and the largest parameter correlation of the joint fit.

Parameters	FHC	RHC	joint (max $\lvert\rho\rvert$)
CC0π	0.147	0.150	0.135
CC0π, π^0	0.186, 0.262	0.168, 0.231	0.152, 0.197 (0.45)
CC0π ν, $\bar{\nu}$	0.159, 0.916	0.169, 0.758	0.143, 0.602 (0.11)
all four classes	0.22, 0.46, 2.10, 0.30	0.20, 0.39, 1.69, 0.26	0.18, 0.31, 1.21, 0.22 (0.69)

6.3 Expected effect on the measurement

The benefit is quantified on the one-bin total, $x = (D - B)H/S$ per mode, with the background B, selected signal S and generated signal per unit flux H taken from the same universes; in a flux universe the integrated flux moves with the generated signal, which reproduces the extraction's flux term (30.1% FHC, against 30.25% in the release). The constraint is the Gaussian posterior of (B, S, H) given the control-region bins, without free parameters (the conditional-covariance constraint of the framework's `ConstrainedCalculator`) or with free class normalisations (generalised least squares).

Table 11: Relative uncertainty (%) of the one-bin total, Asimov, full exposure, FHC / RHC.

	FHC	RHC
Prior total (flux 30.1/27.8, detector 14.6/14.8, GENIE multisim 7.7/7.9, data statistics 4.2/4.1)	35.8	33.6
Conditional constraint from the 2D control-region bins	11.3	11.3
with free CC0π / CC0π+π^0 / all four class normalisations	12.0 / 13.7 / 22.4	11.9 / 13.5 / 23.4
flux not correlated between control and signal regions	31.8	29.6
detector not correlated	18.1	18.2
flux and detector not correlated	34.8	32.8
detector as the unscaled sideband convention (prior 48.0/39.5)	16.0	17.7
Frozen-procedure propagation of a CC0π / CC0π+π^0 fit (item 4(ii))	35.9 / 36.1	33.8 / 34.0

Three things follow. The gain is real but *entirely* a correlated-systematic effect: the control regions constrain the common flux and detector directions of the signal-region prediction, and with those two decorrelated nothing is left. The frozen procedure's propagation rule, which keeps the flux and detector terms and adds the fit uncertainty, gains nothing: a normalisation fit alone cannot reduce the uncertainty of this measurement. Whether it is acceptable to let the control regions constrain the common flux normalisation — the direction the frozen procedure deliberately leaves to the signal-region uncertainty — is a decision for the collaboration, not a technical result. Leave-one-out closure tests how far the Gaussian treatment can be trusted: taking as truth a universe excluded from the covariance, the constrained one-bin total has an RMS bias of 4.5% (prior 26.7%) over 30 flux universes, 10.7% (prior 14.0%) over 30 GENIE universes and 4.6% (prior 5.2%) over the eight knobs, with RMS pulls 0.43, 0.68 and 0.45 and maxima 1.28, 1.64 and 0.99 (FHC; RHC alike). The constraint covers, and it helps against flux and detector variations; it does not constrain the interaction model (Fig. 4).

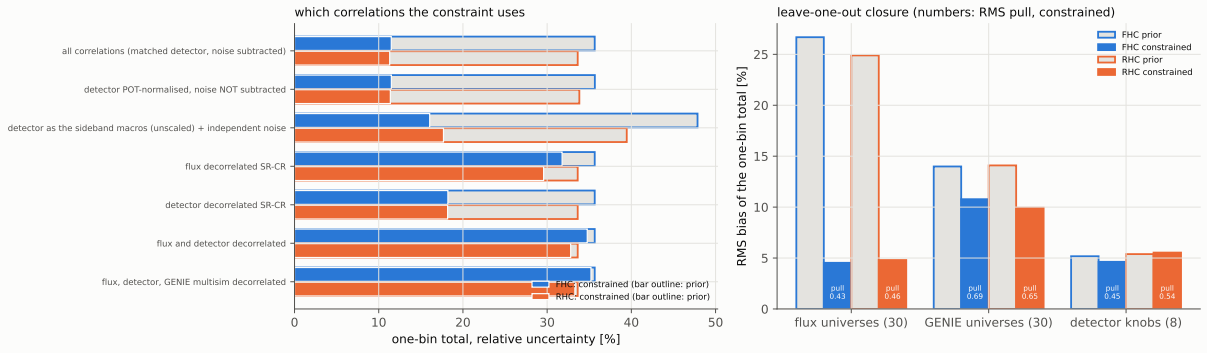


Figure 4: Left: prior (outline) and constrained (filled) one-bin total uncertainty for the covariance variants of Table 11. Right: leave-one-out closure, RMS bias of the constrained one-bin total.

6.4 Validation

The fit is validated entirely on simulation — toy ensembles, injection tests and the convergence of the pull mean — and those studies use no beam-on data, so they are in the analysis note (Sec. 13 of the analysis note) with the rest of the validation programme. The short version is that the fit is unbiased and its uncertainties are correctly sized, with one caveat that changed a default: the combined Neyman–Pearson data variance leaves a small pull-mean bias in the two-dimensional fits, and the Pearson variance at the nominal prediction does not, so the latter

is used.

6.5 Application to the beam-on control regions

Table 12: Fits to the control-region totals of the beam-on data (corrected inputs, corrected FHC Run-1 exposure). LR: likelihood-ratio test against the unfitted prediction. The "no FHC Run 1" rows are unchanged by the exposure correction.

Periods	Model	result	χ^2/ndf (p)
all	no parameters	—	5.73/6 (0.46)
all	CC0π	0.949 ± 0.190 (LR $p = 0.78$)	5.65/5 (0.34)
all	CC0π, π^0	1.097 ± 0.240, 1.346 ± 0.337 (LR $p = 0.57$)	4.61/4 (0.33)
no FHC Run 1	no parameters	—	3.22/6 (0.78)
no FHC Run 1	CC0π	0.984 ± 0.191 (LR $p = 0.93$)	3.22/5 (0.67)
no FHC Run 1	CC0π, π^0	1.059 ± 0.241, 1.171 ± 0.335 (LR $p = 0.88$)	2.97/4 (0.56)
no FHC Run 1	per mode	CC0π, π^0: FHC 1.062, 1.126; RHC 1.046, 1.180 (shared vs. split $p = 0.72$)	
all, per period	no parameters	—	56.00/18 (9×10^{-6})
all, per period	$f_{\mathrm{FHC\,R1}}$	0.869 ± 0.022	18.87/17 (0.34)
all, per period	$f_{\mathrm{FHC\,R1}}$, CC0π, π^0	0.875 ± 0.027, 1.084 ± 0.228, 1.097 ± 0.342	18.73/15 (0.23)

- **No normalisation correction is warranted.** Without FHC Run 1 the unfitted prediction describes the totals, the fitted normalisations are one within their uncertainties, and FHC and RHC agree. The common 10–25% offset is absorbed by the correlated flux and cross-section terms, not by the class parameters.
- **FHC Run 1: the exposure error is resolved; a smaller residual is not.** On the wrong exposure a free factor on the whole FHC Run-1 prediction fitted 0.520 ± 0.020, the quantitative form of the hypothesis of §3. **Resolved**: the beam-on sample is the dataset recorded without the open trigger, 2.192×10^{20} POT with 5 748 692 triggers, confirmed against the beam bookkeeping; the prediction had assumed 3.283×10^{20} and 9 846 635, which put the neutrino MC a factor 0.668 and the beam-off a factor 0.584 too high. The corrected exposure is adopted throughout, the templates and the frozen protocol are re-run on it, and the FHC and combined results are rebuilt on it. With it the frozen global test passes (8.1/8, $p = 0.42$) and the all-period totals fit needs no parameter (5.7/6). **Open**: period by period the data sit above the prediction everywhere except in FHC Run 1 (Table 13); a free Run-1 factor still fits 0.869 ± 0.022 and improves the per-period fit by $\Delta\chi^2 = 37$ for one parameter, while a free factor on any other period gains at most 15. The common 12–25% excess of the other five periods is inside the correlated flux and cross-section terms (which is why the global test passes); a 13 ± 2% offset of one period relative to the others is not covered by anything correlated. Its cause is not identified: the cosmic-enriched region, which is beam-off dominated, sits at 0.95 ± 0.17 in Run 1, so the trigger count is not obviously wrong, and the bookkeeping POT, the completeness of the Run-1 beam-on sample against its run list and the Run-1 overlay normalisation remain to be checked. Until it is understood it is an open item for unblinding, because FHC Run 1 is 28% of the FHC exposure.

Table 13: Control-region totals per period on the corrected exposure, beam-on data over the pre-fit prediction (neutrino MC + beam-off + dirt), from the background-fit templates. Run 2 has no beam-on file.

Period	CC0π	π^0	cosmic
FHC Run 1	$3253/3344 = 0.973$	$1954/1773 = 1.102$	$36/38 = 0.95$
FHC Run 4	$3558/3025 = 1.176$	$2044/1637 = 1.249$	$26/23 = 1.14$
FHC Run 5	$3667/3213 = 1.141$	$2115/1772 = 1.193$	$30/25 = 1.20$
RHC Run 1	$909/752 = 1.209$	$529/483 = 1.094$	$8/10 = 0.83$
RHC Run 3	$6920/6008 = 1.152$	$4706/3768 = 1.249$	$71/47 = 1.51$
RHC Run 4	$4015/3576 = 1.123$	$2620/2176 = 1.204$	$41/33 = 1.25$

- **2D.** Without FHC Run 1 the unfitted prediction gives $\chi^2 = 289.2/86$ in the $(\cos\theta_\mu \times p_\mu)$ bins (all periods 333.0/86 on the corrected exposure; with the stale Monte Carlo it was about 4 900/86). Normalisation parameters do not repair it (276.8/84 with CC0π and π^0 free), and the residual is organised in p_μ: data over the pre-fit prediction is 1.16–1.82 in the 0.35–0.75 GeV/cbins, 1.05–1.40 below 0.35 GeV/cand mostly 0.8–1.1 above 0.75 GeV/c, in every angular bin, both modes and both regions (Fig. 5). This is the p_μ failure of Table 9, it is common to both modes and to regions of different composition, and it points at the muon momentum rather than a background class; it is the open item. 2D normalisation fits are not quoted while it stands.
- **Conditional constraint of the signal-region prediction.** Conditioning the model on the control-region totals (no FHC Run 1, no signal-region data used) moves the expected signal-region total, signal plus background, from 785 ± 211 to 982 ± 83 events (FHC Runs 4, 5) and from $1\,337 \pm 333$ to $1\,648 \pm 138$ (RHC): the 25% offset of the control regions propagated through the correlated terms. It is reported to make the size of that propagation explicit; it is not an input to anything.

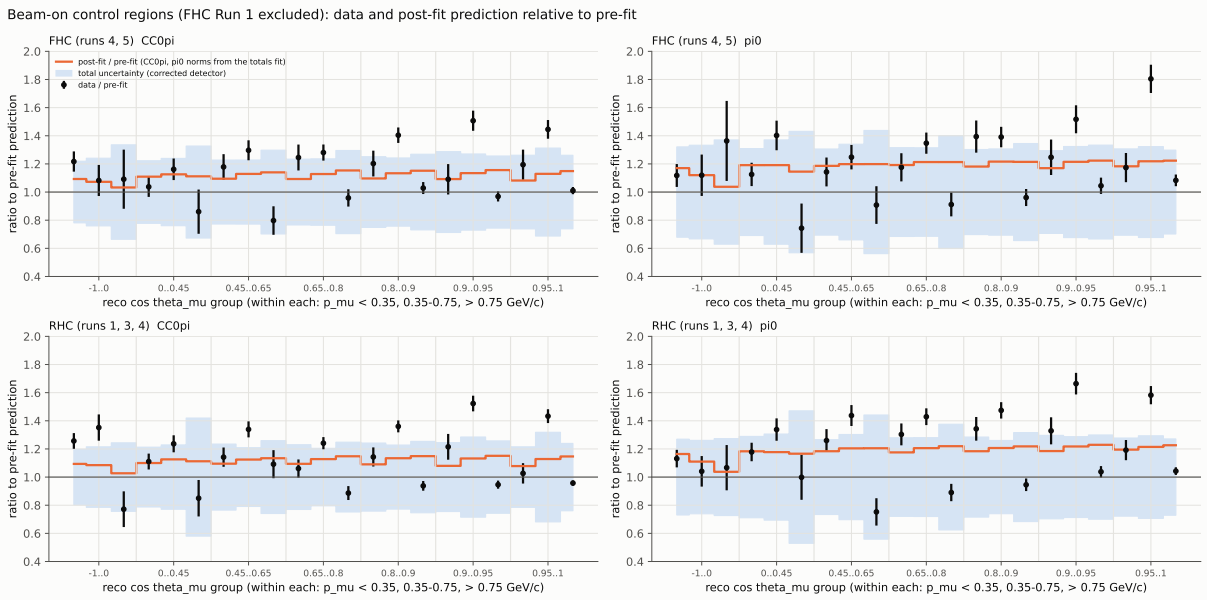


Figure 5: Beam-on control regions without FHC Run 1: data and the post-fit prediction (CC0π and π^0 normalisations from the totals fit) relative to the pre-fit prediction, in reconstructed $\cos\theta_\mu$ groups, each split into $p_\mu < 0.35$, 0.35–0.75 and > 0.75 GeV/c; band: total uncertainty.

7 Far sidebands: control regions with less signal in them

Every control region contains some signal. That is what makes a background constraint bite its own tail: the fit lowers a background component, and the signal region loses the same events a second time through the response. The four regions carry 5–22% signal at data exposure, the

multi-π region worst. A *far* sideband adds one cut that pushes the region away from the signal topology and pays for it in statistics. Three numbers decide whether the trade is worth taking, and Table 14 gives all three: the signal fraction, the events kept, and the background mixture — because a far sideband that constrains a *different* mixture is not a purer version of the near one, it is a different region, and it constrains the signal region less well however clean it is. Nothing here needs approval: these are the control regions already opened, with an extra cut that can only reduce their signal content.

Table 14: Near and far definitions of the four control regions, at data exposure, all eight periods. "signal" is the signal fraction of the total prediction, "kept" the yield relative to the near definition of the same region, and the composition is of the *background* only. Data are the beam-on control-region skim. The composition column is the background only, in the order CC0π / π^0 / multi-π / other / EXT+dirt.

Region	Definition	pred.	data	d/p	signal	composition [%]
CC0π	near (`sb_cc0pi`)	20344	22707	1.12	8.6%	71 / 4 / 2 / 14 / 9
	far: muon is only non-proton	11688	13202	1.13	**2.2**%	85 / 3 / 1 / 10 / 2
	far: + exactly one proton	7889	8941	1.13	**1.7**%	87 / 2 / 0 / 8 / 2
π^0	near (`sb_pi0`)	12001	14383	1.20	9.0%	15 / 51 / 11 / 15 / 9
	far: ≥ 2 showers	6240	7648	1.23	6.3%	8 / 66 / 9 / 11 / 6
	far: ≥ 3 showers	2217	2997	1.35	4.2%	4 / 74 / 8 / 10 / 4
multi-π	near (`sb_multipi`)	220	245	1.12	22.1%	2 / 23 / 56 / 14 / 4
	far: ≥ 4 primary tracks	92	101	1.09	11.9%	2 / 20 / 66 / 11 / 2
cosmic	near (`sb_cosmic`)	175	212	1.21	5.2%	23 / 3 / 4 / 35 / 35
	far: $\theta_{\mu\pi} > 2.9$	105	107	1.02	3.0%	29 / 2 / 2 / 22 / 45
	far: $\theta_{\mu\pi} > 3.0$	80	80	0.99	3.3%	33 / 2 / 1 / 19 / 45

Four things follow.

- **The CC0π region has a cheap far definition.** Requiring the muon to be the only non-proton-like primary track takes the signal fraction from 8.6% to 2.2%, a factor of four, while keeping 58% of the events and *raising* the CC0π content of the background from 72% to 85%. It is purer in both senses at once, which is unusual and makes it the one far definition worth adopting outright. Demanding exactly one proton as well gains little (1.7%) for another third of the statistics.
- **The π^0 region improves more slowly.** Two showers halve the statistics to reach 6.4% signal and three showers cut them to a fifth for 4.3%. The π^0 content does rise to 75%, so the far region is a better-targeted π^0 constraint, but the signal fraction falls more slowly than the events do — the residual signal is CC1$\pi^{\pm}$ with a genuine extra shower, which no shower-counting cut removes.
- **The multi-π region is the contaminated one and cannot be fixed by cuts.** At 22.1% signal it is the region a constraint should trust least, and the only lever available is the track count, which halves it to 12.3% at 42% of the statistics — still four times the signal contamination of the far CC0π region, with 101 data events. The π^0 multiplicity cut is not available here: `pion_number_reco` saturates at two, so "≥ 3 pions" selects nothing, and ≥ 3 non-proton tracks is already implied by the region's own definition.
- **The cosmic region changes character rather than purity.** Signal falls $5.4 \to 3.3$%, but the beam-off and dirt share of the background rises from 33% to 44%. That is the point of the region, so this is a feature; it does mean the far version constrains the cosmic and dirt normalisation specifically rather than the neutrino background.

The most useful number in the table is one that does *not* move. The data-to-prediction ratio is flat under the far cuts — CC0π $1.12 \to 1.13 \to 1.13$, multi-π $1.12 \to 1.09$, π^0 $1.20 \to 1.23$ (the

three-shower version, 1.35 on 4% signal, moves with the shower content, not the signal) — while the signal fraction falls by up to a factor of four. Whatever produces the 12–20% excess of data over prediction in the neutrino-dominated regions, it is therefore not signal contamination and not anything that scales with it. The cosmic region is the exception that locates it: as the far cuts raise its beam-off share from 35% to 45%, its ratio falls from 1.21 to 1.0, so the excess sits in the neutrino simulation, not in the beam-off. (An earlier version of this table quoted an 8–16% *deficit*: the prediction then included the FHC and RHC Run-2 periods, which have simulation but no beam-on file, and the FHC Run-1 exposure was wrong. Both are corrected here.) That is a real constraint on the explanation, and it is available without unblinding anything.

8 Correction found on 2026-10-01: the dirt normalisation

Every prediction in this note took the dirt from the Run-1 dirt sample alone, scaled by the data POT over the summed POT of the neutrino overlays and weighted by 0.65 on top of the central-value weight, which already contains it. The framework scales each run's dirt sample by that run's data POT over the POT of the sample. The placeholder dates from 2026-08-09; it was copied into the control-region macros, the background-fit templates and the far-sideband script, and the extraction never used it. All of them now take the dirt per run from the file list of the release. Over the data periods the dirt rises from 74 to 623 events in CC0π, from 16 to 171 in π^0 and from 2.9 to 44 in the cosmic region; the neutrino simulation and the beam-off are unchanged (Table 15). Tables 16, 17 and 18 replace Tables 9, 12 and 13; Figures 1 and 2 and the background-fit figures are regenerated. The expected uncertainties of Tables 10 and 11 change by less than 0.3% absolute (prior one-bin total 35.6/33.6%, conditional constraint 11.5/11.3%, FHC/RHC).

Table 15: Control-region totals with the corrected dirt, data periods (FHC Runs 1, 4, 5; RHC Runs 1, 3, 4). The dirt used before the correction is in parentheses.

Mode	Region	Data	Prediction	signal	ν bkg	beam-off	dirt
FHC	CC0π	10 660	10 015.7	826.0	8 077.3	840.3	272.1 (40.9)
	multi-π	109	94.9	21.5	69.2	3.9	0.3 (0.1)
	π^0	6 291	5 431.1	482.9	4 399.5	470.7	78.0 (8.6)
	cosmic	92	99.9	4.2	50.3	29.7	15.8 (1.6)
RHC	CC0π	12 047	10 876.6	915.2	8 763.9	846.7	350.9 (33.1)
	multi-π	136	126.1	27.1	94.3	3.6	1.1 (0.1)
	π^0	8 092	6 725.4	594.0	5 578.8	459.7	92.8 (7.0)
	cosmic	120	116.5	5.0	58.0	25.2	28.3 (1.3)

Table 16: Frozen statistics with the corrected dirt; otherwise as Table 9. Shape tests: χ^2/ndf (p); $p < 0.01$ in bold.

Mode	Region	pull (full)	pull (st+det)	σ_{full}/N	p_μ shape	$\cos\theta_\mu$ shape
FHC	CC0π	0.35	3.06	18.4%	13.87/6 (0.031)	6.24/4 (0.182)
	multi-π	0.45	0.81	33.2%	6.42/6 (0.378)	1.69/4 (0.792)
	π^0	0.71	7.08	22.3%	**29.56/6 (< 0.001)**	0.50/4 (0.973)
	cosmic	−0.46	−0.61	17.2%	4.52/6 (0.607)	2.42/4 (0.659)
RHC	CC0π	0.63	4.73	17.2%	**52.45/6 (< 0.001)**	5.14/4 (0.273)
	multi-π	0.25	0.43	31.3%	10.27/6 (0.114)	2.53/4 (0.639)
	π^0	0.97	8.96	20.9%	**66.23/6 (< 0.001)**	1.30/4 (0.861)
	cosmic	0.18	0.24	16.6%	7.71/6 (0.260)	1.11/4 (0.893)

Global normalisation $\chi^2 = 6.03/8$, $p = 0.64$, passes; blind spot of a common shift 50.2%.
Other shape tests all pass: multi-π $\theta_{\mu\pi}$ $p = 0.98/0.99$, cosmic $\theta_{\mu\pi}$ 0.53/0.14, π^0 E_{shr} 0.65/0.62.

Table 17: Fits to the beam-on control-region totals with the corrected dirt; otherwise as Table 12.

Periods	Model	result	χ^2/ndf (p)
all	no parameters	none	4.75/6 (0.58)
all	CC0π	0.975 ± 0.191 (LR $p = 0.89$)	4.73/5 (0.45)
all	CC0π, π^0	1.227 ± 0.242, 1.578 ± 0.341 (LR $p = 0.24$)	1.88/4 (0.76)
no FHC Run 1	no parameters	none	1.95/6 (0.92)
no FHC Run 1	CC0π	1.013 ± 0.192 (LR $p = 0.95$)	1.94/5 (0.86)
no FHC Run 1	CC0π, π^0	1.183 ± 0.243, 1.386 ± 0.339 (LR $p = 0.52$)	0.66/4 (0.96)
no FHC Run 1	per mode	CC0π, π^0: FHC 1.219, 1.436; RHC 1.185, 1.440 (shared vs. split $p = 0.79$)	
all, per period	no parameters	none	61.26/18 (1.3×10^{-6})
all, per period	$f_{\text{FHC R1}}$	0.862 ± 0.021	19.04/17 (0.33)
all, per period	$f_{\text{FHC R1}}$, CC0π, π^0	0.880 ± 0.024, 1.224 ± 0.230, 1.288 ± 0.348	18.04/15 (0.26)

Table 18: Control-region totals per period with the corrected dirt, beam-on data over the pre-fit prediction, from the background-fit templates; otherwise as Table 13.

Period	CC0π	π^0	cosmic
FHC Run 1	3253/3449 = 0.943	1954/1795 = 1.089	36/42 = 0.85
FHC Run 4	3558/3080 = 1.155	2044/1658 = 1.233	26/26 = 0.98
FHC Run 5	3667/3278 = 1.119	2115/1797 = 1.177	30/31 = 0.96
RHC Run 1	909/774 = 1.174	529/489 = 1.081	8/11 = 0.71
RHC Run 3	6920/6192 = 1.118	4706/3817 = 1.233	71/61 = 1.17
RHC Run 4	4015/3673 = 1.093	2620/2198 = 1.192	41/44 = 0.92

Table 19: The cosmic region and its far versions with the corrected dirt, data periods, split into the neutrino simulation, beam-off and dirt (Table 14 gives the background composition).

Definition	ν	beam-off	dirt	pred.	data	d/p
near (`sb_cosmic`)	117.4	54.9	44.0	216.4	212	0.98
far: $\theta_{\mu\pi} > 2.9$	59.3	44.0	32.0	135.3	107	0.79
far: $\theta_{\mu\pi} > 3.0$	45.5	33.4	26.9	105.8	80	0.76

The frozen test passes with more margin, and the RHC cosmic tension (1.60 in the pull, 1.51 in RHC Run 3) was the missing dirt. The fit conclusions stand: no normalisation correction is

warranted and FHC and RHC agree. The FHC Run-1 residual stands as well: a free factor of 0.862 ± 0.021 improves the per-period fit by $\Delta\chi^2 = 42$, against at most 17 for a free factor on any other period, and the other five periods sit 8–23% above the prediction in CC0π and π^0. The far-sideband argument of §7 weakens. The CC0π ratio now rises as the far cuts remove beam-off and dirt (1.09, 1.12, 1.13; π^0 1.18, 1.22, 1.34), and the cosmic region no longer locates the excess in the neutrino simulation: it agrees with the prediction, and its far versions, where the dirt is a quarter of the prediction, lie below it (Table 19). With the neutrino simulation at its central value, the most back-to-back version is described by a dirt scale of 0.0 ± 0.6 (data statistics and the 30% neutrino uncertainty), 1.6 standard deviations below the nominal scale. No dirt normalisation uncertainty enters the control-region prediction or the extraction.

9 Selection change of 2026-10-02: the Bragg-pion requirement

The pion identification of the single-pion selection no longer requires a Bragg-pion score of at least 0.08. The beam-on, beam-off and dirt ntuples of FHC Run 1 and RHC Runs 1 and 3b do not carry the score, so the requirement had applied to the simulation and to only part of the data. An event in which a candidate failed only this requirement now counts that candidate as a pion and leaves CC0π, for the signal region or, with a second pion, the multi-π region. The beam-on files were therefore skimmed again with the new selection and the earlier skims withdrawn. In FHC Run 1 and RHC Runs 1 and 3 the beam-on selection does not change, since the requirement never applied there, and only the prediction moves; in the Run-4 and Run-5 data about 85 predicted events of the new signal region, 6–7% of it, were in the opened control regions before the change. Tables 20, 21, 22, 23 and 24 replace Tables 15, 16, 17, 18 and 19, and the figures are regenerated.

Table 20: Control-region totals with the selection of version 1.7, data periods (FHC Runs 1, 4, 5; RHC Runs 1, 3, 4).

Mode	Region	Data	Prediction	signal	ν bkg	beam-off	dirt
FHC	CC0π	10 563	9 912.7	790.2	8 015.5	835.6	271.4
	multi-π	116	103.8	23.4	75.6	4.5	0.3
	π^0	6 291	5 431.1	482.9	4 399.5	470.7	78.0
	cosmic	95	104.5	4.4	52.0	31.7	16.4
RHC	CC0π	12 004	10 771.5	873.0	8 703.7	843.9	350.9
	multi-π	138	136.2	29.2	100.6	4.1	2.3
	π^0	8 092	6 725.4	594.0	5 578.8	459.7	92.8
	cosmic	123	119.4	5.2	60.2	26.3	27.7

Table 21: Frozen statistics with the selection of version 1.7; otherwise as Table 9. Shape tests: χ^2/ndf (p); $p < 0.01$ in bold.

Mode	Region	pull (full)	pull (st+det)	σ_{full}/N	p_μ shape	$\cos\theta_\mu$ shape
FHC	CC0π	0.36	3.06	18.4%	13.29/6 (0.039)	5.63/4 (0.228)
	multi-π	0.37	0.77	31.5%	7.55/6 (0.273)	1.84/4 (0.766)
	π^0	0.71	7.08	22.3%	**29.56/6 (< 0.001)**	0.50/4 (0.973)
	cosmic	−0.55	−0.74	16.6%	5.36/6 (0.499)	1.65/4 (0.800)
RHC	CC0π	0.67	5.05	17.1%	**50.01/6 (< 0.001)**	5.17/4 (0.271)
	multi-π	0.04	0.08	29.4%	9.69/6 (0.138)	2.33/4 (0.676)
	π^0	0.97	8.96	20.9%	**66.23/6 (< 0.001)**	1.30/4 (0.861)
	cosmic	0.18	0.24	16.6%	7.83/6 (0.251)	1.40/4 (0.844)

Global normalisation $\chi^2 = 7.69/8$, $p = 0.46$, passes; blind spot of a common shift 49.6%.
Other shape tests all pass: multi-π $\theta_{\mu\pi}$ $p = 0.98/0.98$, cosmic $\theta_{\mu\pi}$ 0.59/0.15, π^0 E_{shr} 0.65/0.62.

Table 22: Fits to the beam-on control-region totals with the selection of version 1.7; otherwise as Table 12.

Periods	Model	result	χ^2/ndf (p)
all	no parameters	none	5.58/6 (0.47)
all	CC0π	0.991 ± 0.190 (LR $p = 0.96$)	5.58/5 (0.35)
all	CC0π, π^0	1.250 ± 0.239, 1.605 ± 0.339 (LR $p = 0.21$)	2.41/4 (0.66)
no FHC Run 1	no parameters	none	1.94/6 (0.93)
no FHC Run 1	CC0π	1.021 ± 0.191 (LR $p = 0.91$)	1.92/5 (0.86)
no FHC Run 1	CC0π, π^0	1.203 ± 0.241, 1.419 ± 0.337 (LR $p = 0.46$)	0.39/4 (0.98)
no FHC Run 1	per mode	CC0π, π^0: FHC 1.224, 1.456; RHC 1.206, 1.452 (shared vs. split $p = 0.94$)	
all, per period	no parameters	none	57.66/18 (4.9×10^{-6})
all, per period	$f_{\text{FHC R1}}$	0.869 ± 0.021	19.60/17 (0.30)
all, per period	$f_{\text{FHC R1}}$, CC0π, π^0	0.888 ± 0.023, 1.237 ± 0.228, 1.333 ± 0.344	18.38/15 (0.24)

Table 23: Control-region totals per period with the selection of version 1.7, beam-on data over the pre-fit prediction, from the background-fit templates; otherwise as Table 13.

Period	CC0π	π^0	cosmic
FHC Run 1	3253/3416 = 0.952	1954/1795 = 1.089	36/43 = 0.84
FHC Run 4	3519/3048 = 1.154	2044/1658 = 1.233	27/28 = 0.97
FHC Run 5	3610/3242 = 1.114	2115/1797 = 1.177	32/34 = 0.94
RHC Run 1	909/769 = 1.183	529/489 = 1.081	8/11 = 0.71
RHC Run 3	6920/6129 = 1.129	4706/3817 = 1.233	71/63 = 1.13
RHC Run 4	3972/3638 = 1.092	2620/2198 = 1.192	44/45 = 0.97

Table 24: The cosmic region and its far versions with the selection of version 1.7, data periods; otherwise as Table 19.

Definition	ν	beam-off	dirt	pred.	data	d/p
near (`sb_cosmic`)	121.8	58.0	44.1	223.9	218	0.97
far: $\theta_{\mu\pi} > 2.9$	61.3	47.1	32.1	140.5	110	0.78
far: $\theta_{\mu\pi} > 3.0$	47.4	36.5	27.0	110.8	84	0.76

The conclusions of §8 stand. The frozen test passes and no full-covariance pull exceeds 1; the

p_μ shape failures are those of Table 16. The fits warrant no normalisation correction and FHC and RHC agree. In the older-production periods the CC0π ratio rises by about one percent (FHC Run 1 from 0.943 to 0.952), since the prediction now drops a requirement that their data never had. The FHC Run-1 residual stands: a free factor of 0.869 ± 0.021 improves the per-period fit by $\Delta\chi^2 = 38$, against at most 14 for a free factor on any other period, and the other five periods sit 8–23% above the prediction in CC0π and π^0. The far sidebands do not move beyond their statistics (CC0π 1.09, 1.12, 1.13; π^0 1.18, 1.22, 1.34; cosmic 0.97, 0.78, 0.76), and the most back-to-back cosmic version is still described by a dirt scale of 0.0 ± 0.6.

10 MCS momentum of exiting muons

The muon momentum is the range momentum when the candidate track stops in the detector and the MCS momentum when it exits; 63% of the selected muons exit. The comparisons below use the control-region skim of the beam-on data with the selection of version 1.7, all periods (FHC Runs 1, 4, 5; RHC Runs 1, 3, 4), and the prediction from the overlays, beam-off data and dirt. They use the skim of the two-pion study (`slurm_mp2pi_cr_skim.sbatch`): the events of the single-pion control regions outside the signal regions of both the single-pion and the two-pion selection. The scripts are `cr_mcs_check.py`, `cr_mcs_cause.py` and `mcs_cal_derive.py`.

Observation. The MCS momentum spectrum of exiting candidates has 1.12 times the predicted number of events, the overall normalisation of the regions; its shape falls below the prediction above 1.2 GeV/c (Table 25). The same shape is seen in FHC and RHC, in the old and the new production and in every period. The number of exiting candidates, their visible length and their direction agree with the prediction, and the deficit is present in every length bin. For stopping candidates the ratio of the MCS to the range momentum agrees between data and prediction within 1.5% from 0.15 to 3 GeV/c (233 data events at 1.25–1.5 GeV/c, 91 at 1.5–2 GeV/c), and the range spectrum agrees, including 1.08 above 1.25 GeV/c: the data muons do not have lower momenta, their MCS estimate does. In the released p_μ bins the deficit appears as 0.48 in the highest bin and an excess of up to 1.25 below it.

MCS momentum [GeV/c]	0–0.2	0.2–0.4	0.4–0.6	0.6–0.8	0.8–1.0	1.0–1.2	1.2–1.5	1.5–2	2–3	> 3
data / prediction	1.00	1.18	1.33	1.35	1.23	1.05	0.74	0.38	0.10	0.19
data events	157	1968	4432	5133	4638	3299	2479	1022	109	37

released p_μ bin [GeV/c]	0.15–0.3	0.3–0.425	0.425–0.575	0.575–0.825	0.825–1.25	> 1.25
all candidates	1.00	1.07	1.21	1.25	1.09	0.48
stopping	0.97	0.99	1.04	1.01	1.03	1.08
exiting	1.11	1.20	1.34	1.35	1.11	0.45

Table 25: Exiting muon candidates, MCS momentum spectrum, and the reconstructed muon momentum in the released p_μ bins of version 1.8, beam-on data over the prediction with the normalisation removed, all control regions and periods. Statistical uncertainties of the ratios: 0.01–0.03 (0.09 in the first MCS bin).

Cause. Tracks longer than 1 m are as straight in the data as in the prediction: the angle between the start direction and the chord from start to end is 2.3–3.4° (median per direction bin), 0.07–0.10° smaller in the data, and the chord over the length agrees within 0.001. A coherent bend of the tracks or a lower true momentum would raise that angle. The deficit depends strongly on the drift-direction component of the track and on its length, not on where the track starts or leaves (Table 26). Among the detector variations only the space-charge variation moves the MCS momentum of exiting muons: it lowers the fraction above 1.2 GeV/c to 0.91 of its central value in FHC and in RHC, against 0.49 in the data; every other variation changes it by less than 1%. The data therefore carry an angular noise on the scale of the MCS

segments (14 cm) that the simulation lacks: it adds little to the large scattering of stopping and slow muons, whose fits are anchored by the stopping end, and dominates the small scattering of fast exiting ones. Read as an extra segment-angle spread added in quadrature, the calibration below corresponds to 5–11 mrad, falling slowly with momentum.

$\lvert\hat{d}_x\rvert$ (drift)	bins	0–0.3	0.3–0.6	0.6–0.85	0.85–1		
	ratio	0.69	0.51	0.28	0.43		
$\hat{d}_z$	bins	< 0	0–0.5	0.5–0.8	0.8–0.95	> 0.95	
	ratio	0.51	0.43	0.28	0.56	0.71	
visible length [cm]	bins	0–50	50–100	100–150	150–200	200–300	> 300
	ratio	0.43	0.38	0.37	0.39	0.41	0.72
start x [cm]	bins	0–64	64–128	128–192	192–256		
	ratio	0.58	0.46	0.42	0.48		
exit face	bins	anode	cathode	bottom	top	downstream	
	ratio	0.47	0.48	0.46	0.55	0.49	

Table 26: Fraction of exiting muon candidates with an MCS momentum above 1.2 GeV/c, data over prediction (0.157 over 0.318 overall, ratio 0.49), per bin of the track direction, visible length, start position and the face through which the track leaves. Statistical uncertainties 0.01–0.03 (0.05–0.08 in the sparse bins).

Calibration. In twelve cells of $\lvert\hat{d}_x\rvert$ (edges 0, 0.3, 0.6, 0.85, 1) and visible length (edges 0, 100, 200 cm, open), a quantile map takes the MCS momentum of the simulated exiting candidates to the data. Only candidates matched to a simulated particle are mapped; beam-off events and the cosmic tracks overlaid on the simulation are data already, and the target of the simulated part is the data distribution with their share (9% of the exiting prediction) removed. The map multiplies the MCS momentum by 0.88–1.09 at 0.5 GeV/c and by 0.68–0.87 at 1.5 GeV/c, the strongest change for short tracks with a large drift component. Table 27 gives the closure in the control regions. A parametric map, an extra segment-angle variance and a scale fitted per cell, describes the data less well ($\chi^2 = 169/24$): it cannot keep the high-momentum tail that the data retain, which a noise varying from track to track produces. Together with a coarser and a finer cell grid and maps derived from the old- and new-production data and from FHC and RHC data alone, it provides the uncertainty of the calibration. How the calibration enters the measurement is described in Sec. 12.3 of the analysis note.

Calibration	released p_μ bins						$\lvert\hat{d}_x\rvert$ bins				length bins						χ^2 (MCS)
none	1.00	1.07	1.21	1.25	1.09	0.48	0.69	0.51	0.28	0.43	0.43	0.38	0.37	0.39	0.41	0.72	7937/24
nominal	1.00	1.02	1.04	1.01	0.98	0.97	1.01	1.00	0.99	1.10	0.93	0.98	0.91	1.03	0.80	1.09	27/24
parametric	0.97	0.99	1.07	1.05	0.97	0.94	1.00	0.93	1.16	1.53	1.06	1.12	0.93	1.04	0.72	1.01	169/24
coarse grid	0.99	1.01	1.03	1.00	0.99	0.98	1.13	0.94	1.10	1.26	0.91	0.97	1.01	0.94	1.02	0.99	13/24
fine grid	1.00	1.02	1.04	1.00	0.98	0.98	1.01	1.01	1.00	1.06	0.98	0.97	0.97	0.97	0.99	0.99	24/24
old production	1.00	1.00	1.04	1.00	0.97	1.00	1.08	1.04	1.03	1.28	0.94	1.00	0.98	1.12	0.84	1.12	36/24
new production	1.00	1.03	1.03	1.02	0.98	0.93	0.96	0.96	0.94	1.05	0.93	0.96	0.84	0.94	0.76	1.04	65/24
FHC	1.00	1.03	1.03	1.01	0.99	0.91	0.96	0.94	0.90	1.10	0.93	1.01	0.83	0.93	0.73	1.02	85/24
RHC	1.00	1.00	1.04	1.01	0.97	1.02	1.06	1.06	1.07	1.07	0.94	0.97	0.99	1.13	0.86	1.14	33/24

Table 27: Closure of the calibration in the control regions, all periods: the released p_μ bins of version 1.8 (data over prediction, shape), the fraction of exiting candidates above 1.2 GeV/c per $\lvert\hat{d}_x\rvert$ and length bin as in Table 26, and the χ^2 of the MCS spectrum of exiting candidates (shape). Per period the high-momentum fraction is 0.89–1.06 of the prediction with the nominal map.

Frozen protocol. With the calibrated overlays, dirt and detector variations of version 1.9 (beam-off and beam-on data unchanged) the frozen protocol of §9 passes every shape test. The p_μ tests that failed fall from $\chi^2 = 13.29$, 29.56, 50.01 and 66.23 to 0.84, 3.83, 3.10 and 2.05 for 6 degrees of freedom (FHC CC0π, FHC π^0, RHC CC0π, RHC π^0), and the sum over the eight p_μ tests from 189.5/48 to 20.5/48, against 22.9/48 for the pseudo-data of the calibrated simulation. The region totals, the pulls, the global test ($\chi^2 = 7.69/8$) and the other shape tests do not depend on the muon momentum and are unchanged. The map was derived from these events, all regions and periods together, so this is a closure per region and beam mode.

Hold-out. Maps derived on a random half of the candidates and applied to the other half bring its p_μ bins of version 1.8 to within 6% of the data shape ($\chi^2 = 21.1$ and 5.9 for 5 degrees of freedom, with the halves exchanged), against 7.5 ± 4.0 from the statistics of the derivation and the test alone, estimated with pseudo-data from the calibrated simulation; the MCS spectrum of exiting candidates gives 77 and 103 for 24 degrees of freedom, against 56 ± 22. Maps derived on FHC and tested on RHC, on RHC and tested on FHC, and between the old and the new production move the highest bin by 3–7% relative to the maps from all candidates, as much as the maps from period groups alone move it in Table 27 (0.91–1.02 against 0.97), which make the beam and production terms of the calibration uncertainty (`scripts/mcs_cal_holdout.py`).

11 What this establishes, and what it does not

- The processing chain runs on real beam-on data of both productions, pre-CRT and CRT-era, without error, and the blinding mechanism behaves as designed: no signal-region event was written, and the outputs are refused by the extraction.
- The control-region distributions are complete and are shown for the first time against data.
- **No conclusion about the background model should be drawn yet.** Two bookkeeping issues sit in front of any such conclusion: the missing Run-2 exposure, which is excluded consistently here but leaves 19% of the analysis unexamined, and the FHC Run-1 anomaly, which has the size and the flatness of an exposure error.
- *(Superseded 2026-09-16 and 2026-09-19: with corrected inputs the global test failed because of the wrong FHC Run-1 exposure and passes on the corrected one,* 8.11/8*; the muon-angle failure was an artefact of the stale sideband Monte Carlo; see §5.3 and the status list below.)* The frozen normalisation test passes. The muon-angle shape test does not, in the two high-statistics regions and in both beam modes; the failure survives the removal of the normalisation, is absent from the same code run on pseudo-data, and is largest in RHC, which carries no suspected exposure error. It is the one result here that the bookkeeping issues do not obviously explain, and it is the thing to understand next after them. Its size should not be read from the χ^2: the shape covariance is small once the normalisation is projected out, so a few per cent per bin goes a long way, and the released binning puts three quarters of the $\cos\theta_\mu$ range in one bin.
- The frozen procedure's own response tree treats a single-period excursion as a reconstruction or exposure issue to be investigated, not as a model correction, and forbids excluding a period without independent evidence of a defect. That is the path FHC Run 1 should follow.

Required next.

1. Confirm the exposure and gate count of the staged FHC Run-1 beam-on file against the trigger table, and confirm that the FHC/RHC split of Run 1 at run 6748 is the one the exposure table assumes.

2. Establish whether Run-2 beam-on data exist upstream. If they do, stage and process them; if they do not, Run 2 must be removed from the exposure and the whole normalisation regenerated.

3. Repeat this comparison once both are settled, and only then read the normalisation offset of §3 as a statement about the background.

4. *(Done 2026-09-16: the failure was the stale sideband Monte Carlo, §5.1; the open shape question is now the muon momentum.)* Investigate the muon-angle shape failure independently of the exposure work: it is present in RHC alone, so it can be studied on the RHC periods while the FHC Run-1 question is open. A finer $\cos\theta_\mu$ binning of the control regions, which is not the released signal-region binning and is free to be chosen, would show whether the discrepancy is confined to the forward peak.

5. The prerequisites for opening the signal region are unchanged and none of them is met by this note: the alternative-generator closure, the flux central-value confirmation, the MCS momentum bound, the clean-cache reproduction, and the inverted-PID regions for the misidentification transfers.

Status after 2026-09-16.

1. The muon-angle shape failure is withdrawn: it compared a detector-frame prediction with beam-frame data (§5.1). The p_μ failures of the π^0 region and of RHC CC0π remain and are the open shape question; they point at the muon momentum (§6).

2. With the detector term corrected (§5.2) the frozen global test failed with FHC Run 1 and passed without it; the FHC Run-1 exposure check resolved it (non-open-trigger sample, 2.192×10^{20} POT) and on the corrected exposure the test passes with all periods, 8.11/8 ($p = 0.42$). What remains is the $13 \pm 2\%$ offset of FHC Run 1 relative to the other five periods (Table 13), which is now the gating item.

3. The sideband transfer-factor uncertainties are unaffected by the POT defect; the control-region systematics table of the analysis note (detector column) and the frozen-procedure numbers quoted there are, and must be regenerated with `sb_protocol_data.C` defaults.

4. The joint fit shows that no normalisation correction is warranted and that a normalisation fit propagated under the frozen rules cannot reduce the measurement's uncertainty; a conditional constraint could ($35.8 \to 11.3\%$ on the one-bin total, Asimov), entirely through the flux and detector correlations. Adopting it would change what the control regions are for and needs a collaboration decision.

5. (2026-10-01) The dirt was seven to twenty times too small throughout this note (§8). With it corrected the frozen test passes at 6.03/8 and no correction is warranted; the FHC Run-1 residual (0.862 ± 0.021) remains the gating item; the cosmic far sidebands prefer less dirt than predicted, and the analysis assigns no dirt normalisation uncertainty.

Provenance. Beam-on skim written 2026-09-09 from the eight files of Table 1; prediction from the release of 2026-09-06 (the same Monte Carlo, beam-off and dirt samples, and the same per-run exposure scaling, as the blind studies), with Run 2 removed from both sides. Update of 2026-09-16: sideband Monte Carlo `processed/sb_pi0/`; frozen statistics `macros/sb_protocol_data.C` (defaults: current Monte Carlo, POT-normalised detector variations); templates `macros/bkgfit_templates.C`; detector variations `bkgfit/detvar.py`; fit, validation and data `python3 -m bkgfit.studies` {composition, sensitivity, robustness, validate, ensembles, data}; figures `python3 -m`

`bkgfit.figures`; far sidebands `python3 -m bkgfit.farsb`; logs `logs/bkgfit/`. Update of 2026-10-01: dirt per run (`macros/dirt_perrun.h`, `bkgfit/dirt.py`); frozen statistics `logs/bkgfit/protocol_dirtfix.log`; templates `/data/uboone/processed/bkgfit_dirtfix/`; all other outputs `logs/crdirtfix/`.